Charged-Current Weak Interactions — Part (a) Redo

Full Feynman Integral Approach with Mass Tracking and Fix

Equivalent of Griffiths Eq. (9.7) at High Energy

PHYS 7502 — Particles and Nuclei 2

Problem Statement

(a) Starting with the Feynman Rules for W boson ("Charged Current") interactions (but working at high energy so you can immediately drop *all* mass terms but the W or Z masses) obtain the equivalent of Griffiths Eq. (9.7) without any mass terms inside the traces. **Do NOT go further and evaluate the traces.**

Deliverables

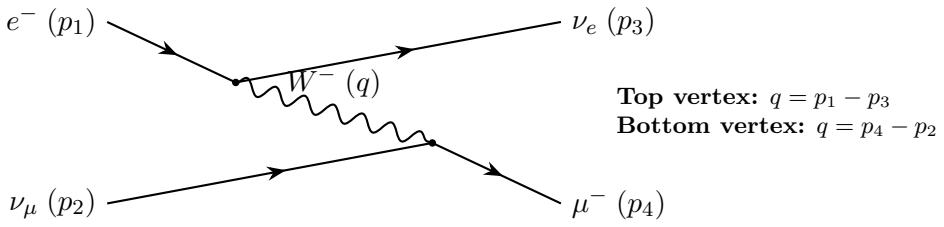
This redo follows your handwritten approach exactly: start from the full Feynman integral form with d^4q and momentum-conserving delta functions, distribute the propagator, and chase every $q_\mu q_\nu$ term down to its mass-suppressed conclusion. **Every step is shown.** One step is fixed mid-derivation — explicitly flagged in a red box — and the rest of the calculation continues to the spin-summed trace form (Griffiths Eq. 9.7 with all fermion masses dropped).

Setup

Conventions. Natural units, $\hbar = c = 1$. Use Griffiths' particle labels for inverse muon decay $\nu_\mu + e^- \to \mu^- + \nu_e$:

$$p_1 = e^-_{\text{in}}, \quad p_2 = \nu_{\mu,\text{in}}, \quad p_3 = \nu_{e,\text{out}}, \quad p_4 = \mu^-_{\text{out}}, \qquad q = p_1 - p_3 = p_4 - p_2.$$

Write $M \equiv M_W$ throughout (only W exchange is involved).



Concepts & Definitions

Feynman rules used (Griffiths §9.1):

- Vertex: $-i\,\dfrac{g_w}{2\sqrt{2}}\,\gamma^\mu(1-\gamma^5)$
- W propagator: $\dfrac{-i\left[g_{\mu\nu} - q_\mu q_\nu/M^2\right]}{q^2 - M^2}$
- Each vertex enforces momentum conservation via $(2\pi)^4\delta^4(\sum p_{\text{in}} - \sum p_{\text{out}})$

- Internal momenta integrated: $\displaystyle\int \frac{d^4q}{(2\pi)^4}$
- Spin sum (Casimir): $\sum_s u(p,s)\bar{u}(p,s) = \not{p} + m$

Step-by-Step Derivation (your approach, every step shown)

Step 1 — Write the full Feynman integral

Following your page-1 setup, multiply: vertex × propagator × vertex, integrate over the internal W momentum, and include the momentum-conserving delta functions at each vertex:

$$\begin{aligned} i\mathcal{M}\,(2\pi)^4\delta^4\big(p_1+p_2-p_3-p_4\big) \;=\; & \int \frac{d^4q}{(2\pi)^4}\,\Big[\bar{u}(3)\big(-i\tfrac{g_w}{2\sqrt{2}}\big)\gamma^\mu(1-\gamma^5)u(1)\Big] \\ & \times\; \frac{-i\big[g_{\mu\nu}-q_\mu q_\nu/M^2\big]}{q^2-M^2} \\ & \times\; \Big[\bar{u}(4)\big(-i\tfrac{g_w}{2\sqrt{2}}\big)\gamma^\nu(1-\gamma^5)u(2)\Big] \\ & \times\; (2\pi)^4\delta^4(p_1-q-p_3)\,(2\pi)^4\delta^4(p_2+q-p_4). \end{aligned}$$

Step 2 — Pull constants outside

Three factors of $(-i)$ from the two vertices and the propagator combine: $(-i)^3 = i$. The two factors of $g_w/(2\sqrt{2})$ multiply to $g_w^2/8$. Pulling these out:

$$\begin{aligned} i\mathcal{M}\,(2\pi)^4\delta^4(p_1+p_2-p_3-p_4) \;=\; & \frac{i\,g_w^2}{8}\int \frac{d^4q}{(2\pi)^4}\,\frac{[\bar{u}(3)\gamma^\mu(1-\gamma^5)u(1)]\,[\bar{u}(4)\gamma^\nu(1-\gamma^5)u(2)]}{q^2-M^2} \\ & \times\; \big[g_{\mu\nu}-q_\mu q_\nu/M^2\big] \\ & \times\; (2\pi)^4\delta^4(p_1-q-p_3)\,(2\pi)^4\delta^4(p_2+q-p_4). \end{aligned}$$

Step 3 — Distribute the propagator into two pieces

Split the bracket: $g_{\mu\nu} - q_\mu q_\nu/M^2$ generates two separate amplitudes added together. Define the spinor currents

$$J_1^\mu \equiv \bar{u}(3)\gamma^\mu(1-\gamma^5)u(1), \qquad J_2^\nu \equiv \bar{u}(4)\gamma^\nu(1-\gamma^5)u(2).$$

Then the propagator-times-currents factor becomes

$$\big[g_{\mu\nu}-q_\mu q_\nu/M^2\big]\,J_1^\mu\,J_2^\nu \;=\; J_1\!\cdot\!J_2 \;-\; \frac{1}{M^2}\,(q_\mu J_1^\mu)(q_\nu J_2^\nu).$$

The two pieces are completely independent algebraic terms. Call them **Term A** (transverse, $g_{\mu\nu}$ piece) and **Term B** (longitudinal, $q_\mu q_\nu$ piece).

Step 4 — Contract the indices

Term A: $g_{\mu\nu}\gamma^\nu = \gamma_\mu$, so

$$\text{Term A} = [\bar{u}(3)\gamma^\mu(1-\gamma^5)u(1)]\,[\bar{u}(4)\gamma_\mu(1-\gamma^5)u(2)].$$

Term B: $q_\mu\gamma^\mu = \not{q}$ inside each current independently:

$$\text{Term B} = -\frac{1}{M^2}\,[\bar{u}(3)\not{q}(1-\gamma^5)u(1)]\,[\bar{u}(4)\not{q}(1-\gamma^5)u(2)].$$

Both q's contract simultaneously — one into each current — because μ and ν are independent dummy indices.

Step 5 — Do the d^4q integral over the first delta function

The d^4q integral is trivial because of the delta function $\delta^4(p_1 - q - p_3)$: it forces $q = p_1 - p_3$ and turns $\int d^4q\, \delta^4(p_1 - q - p_3) = 1$. The remaining $\delta^4(p_2 + q - p_4)$ becomes $\delta^4(p_2 + (p_1 - p_3) - p_4) = \delta^4(p_1 + p_2 - p_3 - p_4)$, which is overall four-momentum conservation. Setting $q^2 \ll M^2$ in the denominator (low-q^2 limit, Griffiths Eq. 9.4 regime), $q^2 - M^2 \to -M^2$. Putting it all together:

$$i\mathcal{M}\,(2\pi)^4\delta^4(\cdots) \;=\; \frac{i\,g_w^2}{8(-M^2)}\,\big[\,\text{Term A} + \text{Term B}\,\big]\,(2\pi)^4\delta^4(p_1 + p_2 - p_3 - p_4).$$

Step 6 — Drop the conservation δ, divide by i to read off $\mathcal{M}$

Cancel $(2\pi)^4\delta^4(\cdots)$ from both sides and divide by i:

$$\mathcal{M} \;=\; \underbrace{\frac{g_w^2}{8M^2}\,\big[\bar{u}(3)\gamma^\mu(1-\gamma^5)u(1)\big]\big[\bar{u}(4)\gamma_\mu(1-\gamma^5)u(2)\big]}_{\mathcal{M}_A\ \text{(want this to survive)}}$$
$$-\;\underbrace{\frac{g_w^2}{8M^4}\,\big[\bar{u}(3)\not{q}(1-\gamma^5)u(1)\big]\big[\bar{u}(4)\not{q}(1-\gamma^5)u(2)\big]}_{\mathcal{M}_B\ \text{(must show this vanishes)}}$$

Step 7 — Reduce $\mathcal{M}_B$ via the Dirac equation (this is where the fix lives)

Fix — What Went Wrong and Why

What I had on page 4 of my work: I substituted $\not{q} = \not{p}_1 - \not{p}_3$ into *both* currents. That works on the top current because $\not{p}_1$ acts on $u(1)$ and $\not{p}_3$ acts on $\bar{u}(3)$ via the Dirac equation. **But on the bottom current neither p_1 nor p_3 matches the spinors $u(2)$ or $\bar{u}(4)$ that are sitting there** — the Dirac equation gives nothing, and the bottom piece sits there unsimplified.
The fix: use momentum conservation. From the diagram, q can equally well be written as $q = p_4 - p_2$ (this comes from the second delta function in Step 5, or just from $p_1 + p_2 = p_3 + p_4$). On the bottom current, use this form instead. Now $\not{p}_4$ is adjacent to $\bar{u}(4)$ and $\not{p}_2$ is adjacent to $u(2)$ — both reduce immediately by the Dirac equation.
Mnemonic: *Always write $\not{q}$ in the form whose momenta match the spinors at that vertex.* Each vertex has its own "natural" form of q — use it.

Top current (use $q = p_1 - p_3$):

$$\bar{u}(3)\,\not{q}\,(1-\gamma^5)u(1) \;=\; \bar{u}(3)\,(\not{p}_1 - \not{p}_3)(1-\gamma^5)u(1).$$

Split into two terms.

First term: $\not{p}_1$ sits to the *left* of $(1-\gamma^5)$, but its partner $u(1)$ sits to the *right*. Anticommute it through using $\{\gamma^\mu, \gamma^5\} = 0$:

$$\not{p}_1(1-\gamma^5) \;=\; \not{p}_1 - \not{p}_1\gamma^5 \;=\; \not{p}_1 + \gamma^5\not{p}_1 \;=\; (1+\gamma^5)\not{p}_1.$$

The $(1-\gamma^5)$ flips to $(1+\gamma^5)$ when commuted past one slash. Now $\not{p}_1$ is adjacent to $u(1)$:

$$\bar{u}(3)\not{p}_1(1-\gamma^5)u(1) \;=\; \bar{u}(3)(1+\gamma^5)\not{p}_1 u(1) \;=\; m_e\,\bar{u}(3)(1+\gamma^5)u(1).$$

Second term: $\not{p}_3$ already sits adjacent to $\bar{u}(3)$. Use the conjugate Dirac equation $\bar{u}(3)\not{p}_3 = m_{\nu_e}\bar{u}(3)$ directly:

$$\bar{u}(3)\not{p}_3(1-\gamma^5)u(1) \;=\; m_{\nu_e}\bar{u}(3)(1-\gamma^5)u(1).$$

Combining:

$$\boxed{\bar{u}(3)\not{q}\,(1-\gamma^5)u(1) \;=\; m_e\,\bar{u}(3)(1+\gamma^5)u(1) \;-\; m_{\nu_e}\,\bar{u}(3)(1-\gamma^5)u(1)}$$

Bottom current (use $q = p_4 - p_2$, the FIX):

$$\bar{u}(4)\not{q}\,(1-\gamma^5)u(2) \;=\; \bar{u}(4)\,(\not{p}_4 - \not{p}_2)(1-\gamma^5)u(2).$$

First term: $\not{p}_4$ already sits adjacent to $\bar{u}(4)$. Use $\bar{u}(4)\not{p}_4 = m_\mu\bar{u}(4)$:

$$\bar{u}(4)\not{p}_4(1-\gamma^5)u(2) \;=\; m_\mu\,\bar{u}(4)(1-\gamma^5)u(2).$$

Second term: anticommute $\not{p}_2$ through $(1-\gamma^5)$ to reach $u(2)$:

$$\bar{u}(4)\not{p}_2(1-\gamma^5)u(2) \;=\; \bar{u}(4)(1+\gamma^5)\not{p}_2 u(2) \;=\; m_{\nu_\mu}\,\bar{u}(4)(1+\gamma^5)u(2).$$

Combining:

$$\boxed{\bar{u}(4)\not{q}\,(1-\gamma^5)u(2) \;=\; m_\mu\,\bar{u}(4)(1-\gamma^5)u(2) \;-\; m_{\nu_\mu}\,\bar{u}(4)(1+\gamma^5)u(2)}$$

Step 8 — Multiply: every term in $\mathcal{M}_B$ carries two fermion masses

$$\begin{aligned}\mathcal{M}_B \;=\; -\frac{g_w^2}{8M^4}\,&\Big[m_e\bar{u}(3)(1+\gamma^5)u(1) - m_{\nu_e}\bar{u}(3)(1-\gamma^5)u(1)\Big]\\ &\times\Big[m_\mu\bar{u}(4)(1-\gamma^5)u(2) - m_{\nu_\mu}\bar{u}(4)(1+\gamma^5)u(2)\Big].\end{aligned}$$

Every cross term in this product is a product of *two* fermion masses (one from the top spinor pair, one from the bottom). Schematically:

$$\mathcal{M}_B \;\sim\; \frac{1}{M^4}\big(m_e m_\mu \;+\; m_e m_{\nu_\mu} \;+\; m_{\nu_e} m_\mu \;+\; m_{\nu_e} m_{\nu_\mu}\big) \times [\text{spinor bilinears}].$$

Step 9 — High-energy limit: drop all fermion masses

Setting $m_e = m_\mu = m_{\nu_e} = m_{\nu_\mu} = 0$ kills every term in $\mathcal{M}_B$ identically. A numerical estimate even at finite mass shows how negligible this is:

$$\frac{\mathcal{M}_B}{\mathcal{M}_A} \;\sim\; \frac{m_e m_\mu}{M_W^2} \;\approx\; \frac{(0.511\,\text{MeV})(105.7\,\text{MeV})}{(80.4\,\text{GeV})^2} \;\approx\; 8\times 10^{-9}.$$

So at high energy:

$$\boxed{\mathcal{M} \;\xrightarrow{E\gg m_f}\; \mathcal{M}_A \;=\; \frac{g_w^2}{8M^2}\,\big[\bar{u}(3)\gamma^\mu(1-\gamma^5)u(1)\big]\big[\bar{u}(4)\gamma_\mu(1-\gamma^5)u(2)\big]}$$

This is exactly Griffiths Eq. (9.6), now derived rigorously by tracking every m_f in $\mathcal{M}_B$.

Physical Insight

One-line interpretation. The $q_\mu q_\nu/M^2$ piece of the propagator is the *longitudinal* polarization mode of the massive W. Longitudinal modes only couple to fermion currents through the fermion mass (this is the Goldstone-equivalence-theorem content at tree level). With light fermions ($m_f \ll M_W$), longitudinal W exchange is suppressed by $m_{f_1} m_{f_2}/M_W^2$ at the amplitude level — once through each vertex, hence the product of two masses.

Step 10 — Compute $\mathcal{M}^*$ via the bar-identity

For each spinor bilinear $[\bar{u}(b)\Gamma\, u(a)]^* = \bar{u}(a)\,\overline{\Gamma}\,u(b)$ with $\overline{\Gamma} \equiv \gamma^0\Gamma^\dagger\gamma^0$. For $\Gamma = \gamma^\mu(1-\gamma^5)$:

$$\Gamma^\dagger = (1-\gamma^5)\gamma^0\gamma^\mu\gamma^0, \quad \overline{\Gamma} = \gamma^0(1-\gamma^5)\gamma^0\gamma^\mu = (1+\gamma^5)\gamma^\mu = \gamma^\mu(1-\gamma^5).$$

The V−A current is its own conjugate. Therefore

$$\mathcal{M}^* \;=\; \frac{g_w^2}{8M^2}\,\big[\bar{u}(1)\gamma^\nu(1-\gamma^5)u(3)\big]\,\big[\bar{u}(2)\gamma_\nu(1-\gamma^5)u(4)\big],$$

with the dummy index relabeled $\mu \to \nu$ to avoid clash.

Step 11 — Form $|\mathcal{M}|^2$, group by fermion line

$$\begin{aligned} |\mathcal{M}|^2 \;=\; \mathcal{M}\mathcal{M}^* \;&=\; \left(\frac{g_w^2}{8M^2}\right)^2 \\ &\times \underbrace{\big[\bar{u}(3)\gamma^\mu(1-\gamma^5)u(1)\big]\,\big[\bar{u}(1)\gamma^\nu(1-\gamma^5)u(3)\big]}_{\text{electron line: } \{1,3\}} \\ &\times \underbrace{\big[\bar{u}(4)\gamma_\mu(1-\gamma^5)u(2)\big]\,\big[\bar{u}(2)\gamma_\nu(1-\gamma^5)u(4)\big]}_{\text{muon line: } \{2,4\}}. \end{aligned}$$

The two fermion lines factor cleanly because they share no spinor.

Step 12 — Spin sum each line, drop m in Casimir

Using $\sum_s u(p,s)\bar{u}(p,s) = \not{p} + m \xrightarrow{E\gg m} \not{p}$, write spinor indices explicitly on the electron line:

$$\begin{aligned} &\sum_{\text{spins}} \big[\bar{u}(3)\gamma^\mu(1-\gamma^5)u(1)\big]\,\big[\bar{u}(1)\gamma^\nu(1-\gamma^5)u(3)\big] \\ &\qquad = [\gamma^\mu(1-\gamma^5)]_{ab}\,(\not{p}_1)_{bc}\,[\gamma^\nu(1-\gamma^5)]_{cd}\,(\not{p}_3)_{da} \\ &\qquad = \mathrm{Tr}\big[\gamma^\mu(1-\gamma^5)\,\not{p}_1\,\gamma^\nu(1-\gamma^5)\,\not{p}_3\big]. \end{aligned}$$

Identical reasoning on the muon line:

$$\sum_{\text{spins}} \big[\bar{u}(4)\gamma_\mu(1-\gamma^5)u(2)\big]\,\big[\bar{u}(2)\gamma_\nu(1-\gamma^5)u(4)\big] = \mathrm{Tr}\big[\gamma_\mu(1-\gamma^5)\,\not{p}_2\,\gamma_\nu(1-\gamma^5)\,\not{p}_4\big].$$

Step 13 — Assemble final result

Key Result

Equivalent of Griffiths Eq. (9.7) with all fermion masses dropped:

$$\sum_{\text{spins}} |\mathcal{M}|^2 = \left(\frac{g_w^2}{8M^2}\right)^2 \mathrm{Tr}\big[\gamma^\mu(1-\gamma^5)\not{p}_1\,\gamma^\nu(1-\gamma^5)\not{p}_3\big] \times \mathrm{Tr}\big[\gamma_\mu(1-\gamma^5)\not{p}_2\,\gamma_\nu(1-\gamma^5)\not{p}_4\big]$$

The traces are left unevaluated, per the problem statement.

Summary of the Fix

Warning / Common Pitfall

What I had on page 4 (close, but stuck):
$\bar{u}(4)(\not{p}_1 - \not{p}_3)(1-\gamma^5)u(2)$ *neither p_1 nor p_3 adjacent to $u(2)/\bar{u}(4)$ — won't simplify*
The corrected step:
$\bar{u}(4)(\not{p}_4 - \not{p}_2)(1-\gamma^5)u(2)$ *both p_4 and p_2 adjacent to their spinors — Dirac equation works*
Why momentum conservation lets this swap:
$q = p_1 - p_3 = p_4 - p_2$ (overall conservation $p_1 + p_2 = p_3 + p_4$). Both forms of $\not{q}$ are mathematically equal; pick the one whose momenta are at the vertex you're simplifying.
Result: both currents reduce to $m \times$ (spinor bilinear), giving $\mathcal{M}_B \propto m_e m_\mu / M_W^2 \to 0$ at high energy.

Intuition

Why the symmetric choice failed. The Dirac equation collapses $\not{p}_k\, u(p_k) = m_k u(p_k)$ *only* when the slashed momentum matches the spinor's momentum. If you write $\not{q}$ in a form whose momenta don't appear at the vertex you're working with, the slash is just a 4×4 matrix floating in the middle of an inert spinor sandwich — no simplification possible. **Each vertex has its own "natural" decomposition of q.** Use it.

Warning / Common Pitfall

Bookkeeping note for the rest of the calculation. The final $\sum_{\text{spins}} |\mathcal{M}|^2$ above is the un-averaged sum. To get the cross section you'll later divide by the initial-state spin multiplicity — which for $\nu_\mu e^- \to \mu^- \nu_e$ is just 2 (two electron spin states; the incoming neutrino has only one because $(1-\gamma^5)$ kills its right-handed component). Griffiths applies this $\frac{1}{2}$ later, between Eq. (9.10) and Eq. (9.11).